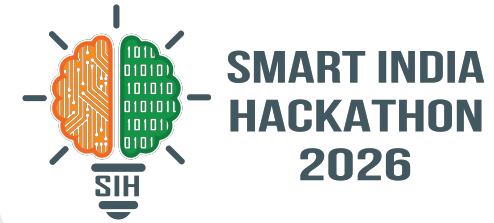
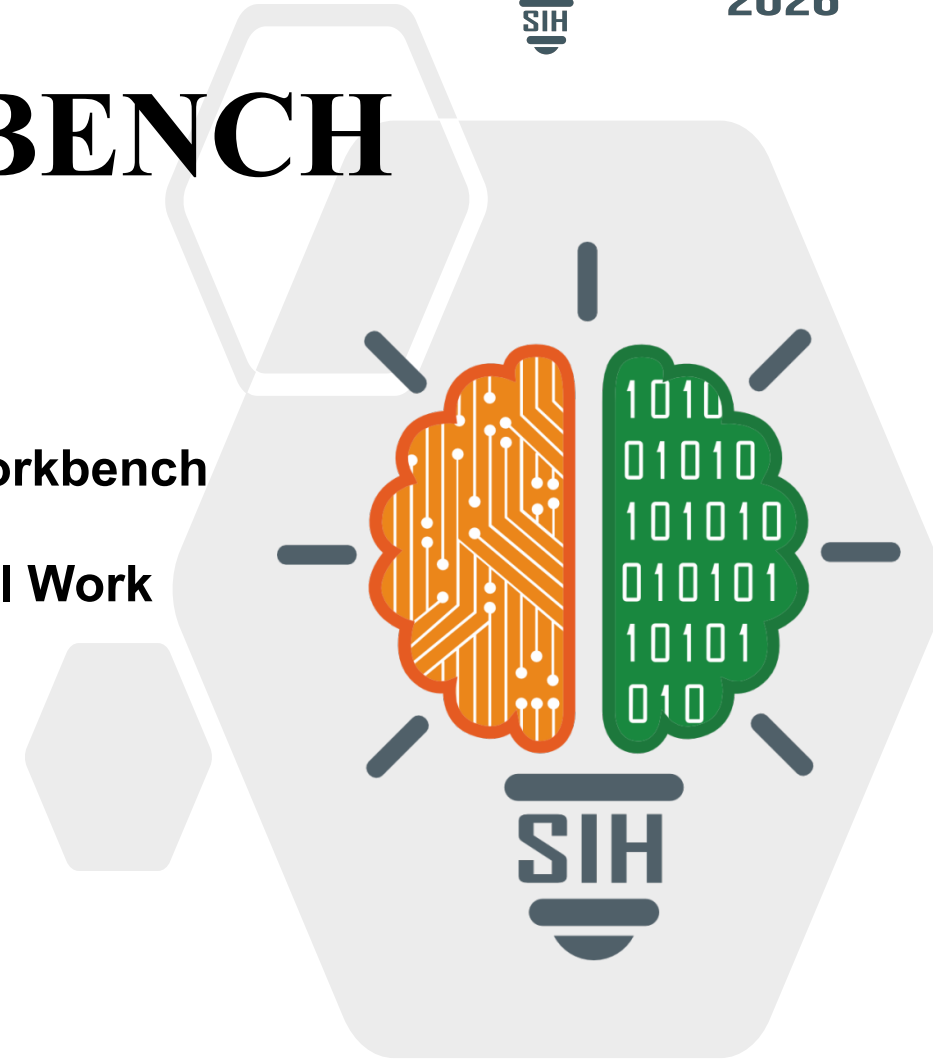


SMART INDIA HACKATHON 2026



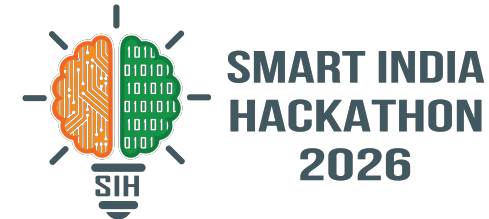
SOVEREIGN AI WORKBENCH

- **Problem Statement ID : 26117**
- **Problem Statement Title : Sovereign On-Premise Agentic AI Workbench using Open-Weight Multimodal LLMs for Confidential Industrial Work**
- **Theme : Smart Automation**
- **PS Category : Software**
- **Team ID : TBD**
- **Team Name : Tech Byte**





Proposed Solution Prototype



A Secure, hosted AI workbench that enables organizations to use powerful AI on **confidential industrial data** without sending data outside their infrastructure.



Your Data
Your Infrastructure



Open-Source &
Cost-Effective



Domain-Specific
Intelligence



Productivity
At Scale



Secure, Compliant
& Future-Ready



1. SECURE & PRIVATE

KEEP YOUR DATA IN YOUR CONTROL

- ✓ On-premise/ air-gapped deployment
- ✓ Data stays inside the organisation
- ✓ No external AI / Cloud calls
- ✓ Role-based access control (RBAC)
- ✓ Secure access, audit logs & monitoring
- ✓ Designed for enterprise and government use



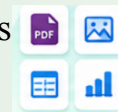
YOUR DATA | YOUR INFRASTRUCTURE | YOUR CONTROL



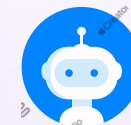
2. INTELLIGENT & MULTIMODAL

UNDERSTAND ANY TYPE OF DATA

- ✓ Multiple open-weight LLMs (e.g, Llama, Qwen)
- ✓ Automatic model selection based on task
- ✓ Understand PDFs, scans, images, drawings
- ✓ Local OCR + Vision + RAG
- ✓ Works with domain-specific knowledge base
- ✓ Provides accurate, cited and verifiable answers



TEXT | IMAGES | TABLES | DRAWINGS | MULTIMODAL AI



3. AGENTIC & ACTION-ORIENTED

TURN INSIGHTS INTO ACTIONS

- ✓ Plans and executes multi-step tasks
- ✓ Uses local tools for files, code & spreadsheets
- ✓ Searches internal knowledge (RAG)
- ✓ Creates Words, PPT, Excel, reports & code
- ✓ Automates routine and complex workflows
- ✓ Human-in-the-loop for critical actions
- ✓ Scalable and customizable for different use case



SEARCH | ANALYZE | ACT | AUTOMATE



Confidential &
Secure Operation



Higher
Productivity



Faster & Better
Decision Making



Reduced
Manual Effort

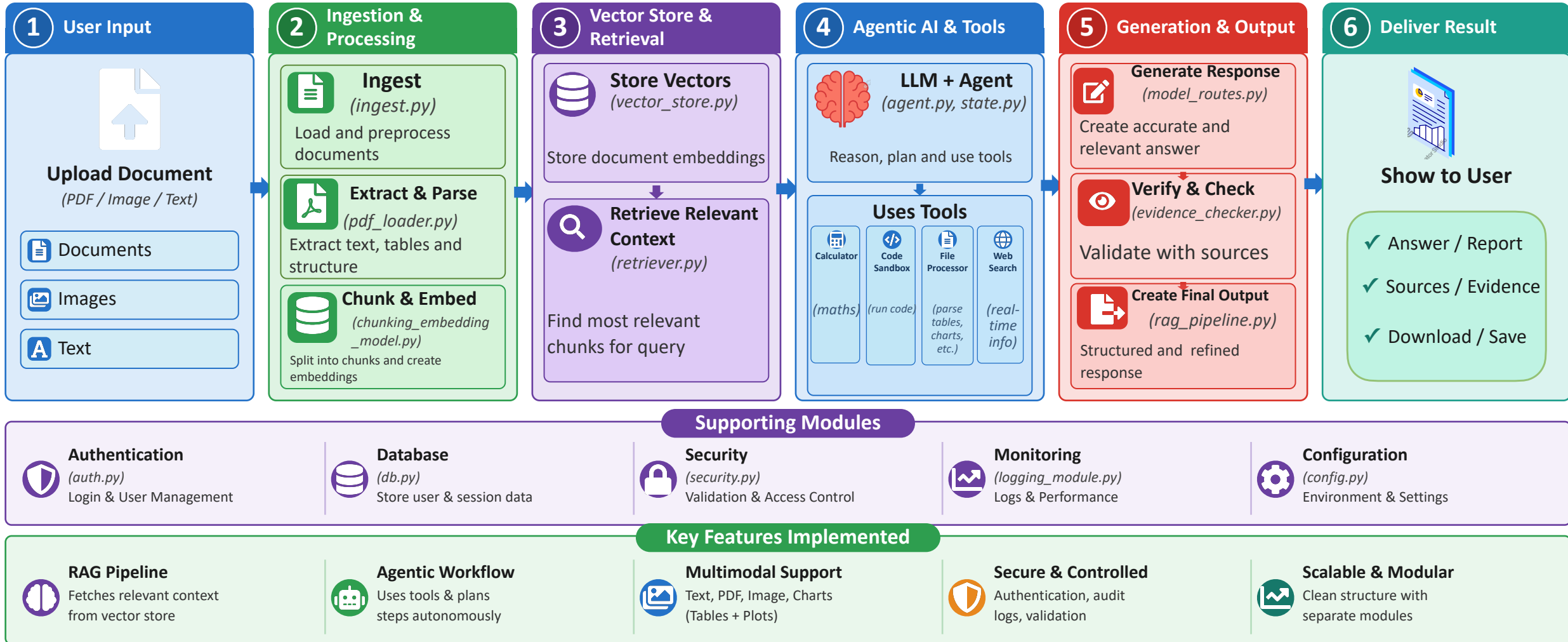


Cost-Effective
& Sustainable



Scale for Use
Enterprise

BRING AI TO YOUR DATA - NOT YOUR DATA TO AI





CAN THIS WORK IN A REAL INDUSTRIAL ENVIRONMENT ?

Yes — with the right architecture, safeguards and domain adaptation, our solution is practical, scalable and ready for real-world use.



**REAL PROBLEMS
REAL SOLUTIONS**



WHAT WE HAVE BUILT

KEY COMPONENTS OF OUR SOLUTION



Local Open-Weight Models

(Llama, LLaVA, Qwen)
Runs locally, no dependency on closed APIs



Agentic AI + Model Router

(Task-based selection)
Automatically chooses the right model & tools



RAG + FAISS Knowledge Base

(Company documents)
Uses internal data for accurate answers



OCR + Vision Processing

(PDFs, images, drawings)
Understands text, tables, charts and diagrams



Local Tools & Automation

(Files, Spreadsheets, Python)
Performs calculations, analysis and automation



Security & Access Control

(JWT + RBAC + Audit)
Enterprise-grade security and data isolation



Python Sandbox

(Safe code execution)
Executes analysis in a controlled environment



KEY CHALLENGES

REAL-WORLD CONSTRAINTS



GPU Limitations

Limited compute resources for large models, especially in on-premise environments.



Complex Documents

Scanned PDFs, tables, drawings, handwritten content and multi-format data.



AI Hallucination

Incorrect or unsupported information.



Sensitive Data

Confidential industrial information must stay private and secure.



Code Execution Safety

Risk of malicious or harmful code execution.



Integration & Scalability

Working within existing IT systems and scaling for real users.



OUR MITIGATION STRATEGIES

HOW WE SOLVE IT



Smaller / Task-Specific Models

Optimized for local GPU/CPU and specific tasks (model routing, quantization).



OCR + Vision + RAG

Extract, understand and retrieve relevant information from complex documents.



RAG + Evidence Check + Human Review

Grounded responses with source references and verification.



On-Premise + RBAC

Keep data local with role-based access control, encryption and audit logs.



Restricted Sandbox

Safe and controlled code execution environment (with monitoring and limits).



Modular & Scalable Architecture

Containerized deployment, easy integration with enterprise systems, and support for future scaling.



**FROM INDUSTRIAL CONSTRAINTS TO
DEPLOYABLE AI →**



**Technically
Feasible**



**Practically
Viable**



**Secure &
Responsible**



**Scalable for
Real-World Impact**

→ TURNING SECURE AI INTO REAL-WORLD IMPACT

Secure Data | Faster Work | Smarter Decision | Measurable Benefits



POTENTIAL IMPACT ON TARGET AUDIENCE



ENGINEERS & TECHNICAL TEAMS

WORK SMARTER NOT HARDER

Save Time
Increase Productivity

- ✓ Faster document & drawing analysis
- ✓ Instant SOP / manual search
- ✓ Quick access to standards & guidelines
- ✓ Reduce repetitive work
- ✓ Focus on design analysis and innovation



MANAGEMENT AND DECISIONS MAKERS

FASTER DECISIONS GREATER CONTROL

From Information to Action

- ✓ Faster approval and reviews
- ✓ AI-powered insights from documents
- ✓ Automated report generation
- ✓ Data-driven decision support
- ✓ Better compliance & governance
- ✓ Enhanced transparency



ORGANIZATION / ENTERPRISE

SECURE + COMPLIANT + FUTURE READY

- ✓ Data sovereignty / (keeps data on-premise)
- ✓ Local AI infrastructure
- ✓ Process automation & efficiency
- ✓ Regulatory compliance (RBAC audit logs)
- ✓ Scalable for enterprise use
- ✓ Supports sustainable & paperless operations
- ✓ Knowledge retention and easy access

Secure Today
Stronger Tomorrow

KEY BENEFITS OF OUR



PRIVACY & SECURITY

Confidential data stays on premise

TRUSTED & COMPLIANT



PRODUCTIVITY

Faster knowledge work & analysis

MORE WORK IN LESS TIME



COST EFFICIENCY

Less reliance on external tools

OPTIMIZED RESOURCE



AUTOMATION

Multi - step tasks Handled by AI agents

REDUCED MANUAL EFFORT



KNOWLEDGE ACCESS

Organizational Knowledge at your fingertips (RAG)

UNIFIED KNOWLEDGE HUB



ENVIRONMENT

Reduced paper-based workflow

GREENER OPERATIONS



CONFIDENTIAL DATA



SECURE AI (On-Premise)



RAG + AGENTS (Knowledge + Action)



FASTER WORK



SMARTER DECISIONS



REAL-WORLD IMPACT

RESEARCH FOUNDATION • DOCUMENTATION • SECURITY

Built on Trusted Research, Proven Frameworks and Industry Best Practices

1 RAG

Retrieval-Augmented Generation



- ✓ Ground responses with trusted data
- ✓ Reduces hallucinations
- ✓ Supports domain-specific knowledge
- ✓ Ideal for enterprise & government use

[Research Paper](#)

2 ReAct

Reasoning + Acting with Tools



- ✓ Combines reasoning and action
- ✓ Uses external tools (search, calculator, DB)
- ✓ Improves factuality and reliability
- ✓ Effective for complex, multi-step tasks

[Research Paper](#)

3 LangGraph

Agent Orchestration, Workflows & Routing



- ✓ Build complex agent workflows
- ✓ State management and memory
- ✓ Tool routing and human-in-the-loop
- ✓ Scalable and production-ready agents

[Research Paper](#)

4 Ollama

Local LLM Runtime for On-Premise AI



- ✓ Run open-source models locally
- ✓ Supports Llama, LLaVA, Qwen and more
- ✓ No dependency on closed APIs
- ✓ Ensures data privacy and control

[Official Documentation](#)

5 FAISS

Vector Similarity Search & Retrieval



- ✓ High-performance vector search
- ✓ Enables fast and accurate retrieval
- ✓ Works with large-scale embeddings
- ✓ Widely adopted in RAG pipelines

[Official Document](#)

6 NIST AI RMF

AI Risk Management Framework



- ✓ Guidelines for trustworthy AI
- ✓ Risk identification and mitigation
- ✓ Fairness, safety, privacy and security
- ✓ Helps in responsible & compliant deployment

[Official Guideline](#)